Exam Practice: Mixed Review

Year 5 Accelerated · **Homework Worksheet**

Worked Example

Mixed Practice: A sum of £150 is shared in the ratio 2:3. Find 20% of the larger share.

Answer: Larger share = 3/5 × £150 = £90; 20% of £90 = £18.

Mixed Core (15 questions)

1. Write 6,000 + 300 + 40 + 2 as a single number.
2. Calculate 4,562 + 3,879.
3. Calculate 8,001 − 2,647.
4. What is 27 × 8? A) 196 B) 206 C) 216 D) 226 E) 236
5. Calculate 7,296 ÷ 6.
6. What is 5/9 of 81?
7. Calculate 9.4 − 3.75. A) 5.55 B) 5.65 C) 5.75 D) 5.85 E) 5.95
8. Find 35% of 180.
9. Share £96 in the ratio 5:3. What is the smaller share?
10. If 4y − 5 = 23, what is the value of y? A) 5 B) 6 C) 7 D) 8 E) 9
11. How many degrees are there in a full turn?
12. Convert 3,400 ml into litres.
13. Find the median of 3, 9, 5, 11, 7. A) 5 B) 6 C) 7 D) 9 E) 11
14. Select ALL that apply: which of these numbers are factors of 48? A) 6 B) 9 C) 12 D) 14 E) 16
15. A lorry travels at a steady 55 mph for 3 hours. How far does it travel?

Mixed Applied (15 questions)

1. A stationery set costs £3.60. Priya buys 4 sets. How much change does she get from a £20 note?
2. A recipe uses 180 g of sugar to make 12 muffins. How much sugar is needed for 20 muffins?
3. A train covers 210 miles at an average speed of 70 mph. How long does the journey take? A) 2.5 hours B) 3 hours C) 3.5 hours D) 4 hours E) 4.5 hours
4. In a school of 450 pupils, 60% are girls. How many boys are there?
5. A rectangular field is 24 m long and 9 m wide. What is its perimeter?
6. Jack has £40. He spends 3/8 of it on a game. How much does he have left?
7. A £60 bicycle helmet is reduced by 15% in a sale. What is the sale price? A) £49 B) £50 C) £51 D) £52 E) £53
8. Four friends share a £52 taxi fare equally. How much does each person pay?
9. A bottle holds 2 litres of water. 450 ml is drunk. How much water remains, in millilitres?

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10. Select ALL that apply: a box contains yellow, pink and white sweets in the ratio 4:1:5. Which statements are true? A) White sweets are the most common colour. B) For every 10 sweets, 4 are yellow. C) Pink sweets make up 1/10 of the box. D) Yellow sweets make up half of the box. E) There are fewer pink sweets than yellow sweets.

11. A digital clock shows 8:25 am. What time will it show in 50 minutes?

12. A film starts at 6:45 pm and lasts 2 hours 20 minutes, including trailers. At what time does it finish?

13. A number is doubled and then 9 is added, giving 37. What was the original number? A) 12 B) 13 C) 14 D) 15 E) 16

14. A rectangular patio measuring 10 m by 6 m is to be paved. Paving costs £15 per square metre. What is the total cost?

15. A car's tank holds 45 litres and the car uses fuel at 9 litres per 100 km. How far can it travel on a full tank?

Standard / Advanced (15 questions)

1. What is 40% of half of 90?

2. A number is one third of 72. What is double that number?

3. What is 15 less than three times 20?

4. Find the mean of the first four even numbers.

5. What is the remainder when 7 times 13 is divided by 5?

6. A rectangular lawn measures 14 m by 9 m. A path 1 m wide is built all the way round the inside edge of the lawn. What is the area of the remaining grass?

7. A shop buys 60 toys for £180 in total, then sells each toy for £4.20. What is the shop's total profit? A) £62 B) £67 C) £72 D) £77 E) £82

8. Ella scores 68, 74 and 81 in three tests out of 100. What mark out of 100 does she need in a fourth test to have an average of 75 across all four tests?

9. A tank contains 240 litres of water. Water is drained at 8 litres per minute for the first 10 minutes, then at 5 litres per minute after that. How long in total does it take to empty the tank? A) 36 minutes B) 38 minutes C) 40 minutes D) 42 minutes E) 44 minutes

10. A map scale is 1 : 50,000. A field is drawn as a rectangle 3 cm by 2 cm on the map. What is the actual area of the field, in hectares? (1 hectare = 10,000 m^2)

11. Three numbers are in the ratio 2:3:4. The largest number is 36 more than the smallest. Find the sum of the three numbers. A) 144 B) 150 C) 156 D) 162 E) 168

12. Select ALL that apply: a school trip costs £18 per pupil. 92 pupils go on the trip, and the school also pays a flat £150 coach hire fee on top. Which statements about the total cost are true? A) The pupil payments alone total £1,656. B) The total cost including coach hire is £1,806. C) The average cost per pupil, including coach hire, is more than £19. D) If 8 more pupils joined at the same rate, the pupil payments alone would total £1,800. E) The coach hire fee is more than 10% of the pupil payments.

13. A cyclist sets off at 09:15 and rides at an average speed of 24 km/h. She covers 54 km in total, stopping for a 20-minute break exactly halfway through her ride. At what time does she finish the ride?

14. A number, when increased by 25%, becomes 60. What is 10% of the original number?

15. A cube has a volume of 125 cm^3. What is the total surface area of the cube?

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